

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



May 7, 2007

MATERIALS LABORATORY FACTUAL REPORT

Report No. 07-064

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : I-90 Connector Tunnel to the Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. TESTS CONDUCTED

Lap shear strength tests and lap shear creep tests were conducted on Powers Fasteners Power-Fast Standard Set and Fast Set epoxies.

C. DETAILS OF THE TESTING

Powers Fasteners¹ Power-Fast Standard Set and Fast Set epoxy anchor systems were sent to an outside laboratory² for testing to investigate their creep behavior in shear and to determine their shear strengths. Reports documenting the testing are attached.

The Safety Board provided dual cartridges of resin and hardener along with a manual injection tool and static mixer nozzles, and test samples were manufactured at the testing laboratory. All of the Standard Set samples were made using a cartridge marked with lot number 062700H2LC and use by date of 09/2008. The Fast Set cartridge used for the creep testing was marked with lot number 062570I1LC and use by date of 09/2008. The Fast Set cartridge used for the shear strength testing was marked with lot number 061940I1LC and use by date of 06/2008. The lot number indicates the date of packaging; the first two digits indicate the year and the next three numbers indicate the day (out of 365 days), so 06270 = September 27, 2006.

The shear strengths of the epoxies were measured per ASTM D 5656³. The shear creep tests were carried out using tension loading on samples constructed as specified by ASTM D 5656.

¹ Powers Fasteners, Brewster, New York.

² Bodycote Polymer – Broutman Laboratory, Melrose Park, Illinois.

³ ASTM D 5656, Standard Test Method for Thick-Adherend Metal Lap-Shear Joints for Determination of the Stress-Strain Behavior of Adhesives in Shear by Tension Loading, ASTM International, West Conshohocken, Pennsylvania.

The bonded area of the lap shear specimens used for both the creep and strength testing was nominally 1 inch by 1 inch. Bond thicknesses were chosen to match a gap of 0.0625 inch between 0.625-inch diameter anchors and a 0.75-inch diameter hole in the concrete, as was installed in the I-90 Connector Tunnel.

The initial target stress levels and testing environments for the creep tests were based on the approximate level of anchor loading in the I-90 tunnel and on the loads and environments specified in test protocols found in ASTM E 1512⁴ and the ICC-ES Acceptance Criteria AC58⁵.

Loads on each anchor in the tunnel were estimated to be on the order of 2000 pounds. Assuming that load is spread evenly over an area of 10.80 square inches (a cylindrical surface 0.6875 inch in diameter and a 5-inch embedded anchor depth, the average shear stress for a 2000 pound load would be 185 psi.

The anchor creep test in ASTM E 1512 and ICC-ES AC58 specify a load that is 40 percent of the ultimate strength of the anchor. Powers Fasteners Specification and Design Manual lists an ultimate strength of 23,400 pounds for Standard Set epoxy using 0.625-inch diameter anchors embedded to a depth of 5.626 inches in 4,000 psi concrete. A footnote indicates that the ultimate load of the Fast Set epoxy should be reduced from the ultimate load of the Standard Set epoxy by 25 percent. Assuming a cylindrical surface of 0.6875 inch in diameter and 5.625 inches long gives an area of 12.15 square inches. A load of 23,400 pounds distributed evenly over that area gives an average shear stress of 1926 psi, and 40 percent of that stress is 770 psi. This stress level was used for both epoxy materials, so the Fast Set epoxy would be tested at approximately 50 percent of the ultimate strength reported by Powers Fasteners. ASTM E 1512 and ICC-ES AC58 also specify that the creep test be performed at 110 °F.

Four different initial test parameters were specified for the creep testing of both the Standard Set and Fast Set epoxies. The initial test parameters were chosen in a effort to assure that there would be significant measured creep within the 42-day testing window. A test in wet conditions was included to investigate whether moisture could have played a role in the ceiling collapse. The initial parameters were:

1. 770 psi, 70 °F, dry
2. 770 psi, 70 °F, wet
3. 770 psi, 110 °F, dry
4. 185 psi, 110 °F, dry

Specimen failures during the initial load application required a revision of some test conditions to use lower applied stress levels.

The lap shear strength tests were carried out at 72 °F. The mean shear strength of the Standard Set epoxy was determined to be 2120 psi with a standard deviation of the mean of 50 psi. The mean shear strength of the Fast Set epoxy was determined to be 710

⁴ ASTM E 1512, Standard Test Methods for Testing Bond Performance of Bonded Anchors, ASTM International, West Conshohocken, Pennsylvania.

⁵ ICC-ES AC58, Acceptance Criteria for Adhesive Anchors in Masonry Elements, ICC Evaluation Service, Inc., Whittier, California.

psi with a standard deviation of the mean of 120 psi. Powers Fasteners Specification and Design Manual lists the slant shear strength per ASTM D 732⁶ for the Standard Set and Fast Set epoxy anchor systems. This ASTM test method designation might be an error, as ASTM D 732 covers a method to determine the shear strength by punching a hole through a sample, whereas ASTM C 882⁷ covers slant shear testing of concrete samples bonded by epoxy adhesives. The Standard Set slant shear strength is listed as 6,700 psi (tested after 14 days) and the Fast Set slant shear strength is listed as 4,900 psi (tested after 1 day).

Carl R. Schultheisz
Materials Research Engineer

⁶ ASTM D 732, Standard Test Method for Shear Strength of Plastics by Punch Tool, ASTM International, West Conshohocken, Pennsylvania.

⁷ ASTM C 882, Standard Test Method of Bond Strength of Epoxy-Resin Systems Used With Concrete By Slant Shear, ASTM International, West Conshohocken, Pennsylvania.

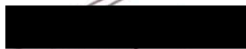
30003712

NTSB

Lap Shear Creep Testing of Epoxy Adhesives

Date: 27th March 2007

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INDEX

REPORT

APPENDIX

Introduction

Bodycote Polymer – Broutman Laboratory (BPBL) was contracted by National Transportation Safety Board (NTSB) to conduct lap shear creep tests on two types of epoxy adhesives. NTSB specified the stress levels and the temperatures used for the creep tests. The epoxy adhesive used in the study was the Power-Fast + Epoxy Adhesive system (Power Fasteners, Brewster, NY). It is a two-component structural epoxy, which is designed for use in anchoring threaded rods, bolts, reinforcing bars, and smooth dowels into concrete and masonry. Power-Fast is available in standard set and fast set formulas. NTSB supplied the Power-Fast plus Standard Set Epoxy, Power-Fast plus Fast Set Epoxy, manual dispensing tool and the mixing nozzles.

Sample Preparation

Test specimens of the required dimensions were prepared. Rectangular sections of aluminum were milled to the required length. The aluminum blanks (6061) and the shims were then cleaned using trichlorethane to remove any grease. Then the specimens were washed with deionized water. After it is dried, BR 127 primer was applied to all sides of the specimen. Care was taken for those specimens in which the creep tests were conducted in wet conditions. Then the primer was cured at 250°F for 30 minutes. The aluminum sections were bonded using the epoxy adhesives and allowed to cure as per the data sheet supplied by the manufacturer. After the specimens have fully cured, they are machined to get the required dimensions. The test specimen geometry is shown in Figure1.

Creep Test Set up

Creep test stands have to be adjusted to corresponding stress levels before performing the tests. The required weights used for the test were determined based on the stress level on the specimen and the specimen dimensions. Table 1 shows the test matrix for performing the creep tests. Once the loads have been decided, the creep stands was adjusted for the corresponding loads. The following below list the procedure that was carried out. At first load cell was attached to the creep stand as shown in Figure 2. Then the creep stand was balanced under no load. To perform this operation, the level was mounted on top of the stand and the nut provided is adjusted until the creep stand is balanced. Once the machine

was balanced, required loads as per the calculation were added to the creep stand. The digital read out connected to the load cell was used to read the exact load values. The creep stands have 10:1 ratio arm. So, for every 10 pounds of load added, it will reflect 100 pounds on the specimen. The load was removed once the creep stands were calibrated. Dial gauges to read the displacements were mounted on the rod using special attachment as shown in Figure 3. The specimen was mounted on the stand and the machine is balanced again using the nut provided in the creep stands. For high temperature setup, a furnace was fitted to the creep stand and temperature was acquired using Daqbook data acquisition (Omega.) as shown in Figure 4. The lap shear creep test setup is shown in Figure 5.

Results and Discussions

Creep test was carried out on both standard and fast set epoxies. The results of the creep tests were shown in Appendix. Standard set epoxy sample loaded at 770 psi, 110°F, dry condition failed while loading. The failure surface is shown in Figure 6. From the figure, it can be clearly seen that the failure is cohesive failure of the adhesives. Standard set epoxy sample loaded at 770 psi, 70°F, wet condition failed at 988 hours. The failure surface is shown in Figure 7. Fast set epoxy samples failed at higher stress levels and elevated temperature (770 psi and 110°F). The failure occurred during loading the specimens in the creep stand. Also it can be seen from Figures 8-13, the failure is cohesive in nature. Fast set epoxy tested at lower stress levels (385 psi) showed cohesive failure at 385 psi for both dry and wet condition.

Table 1 Test Matrix for the lap shear creep tests on standard set and fast set epoxy adhesives.

S.no	Type	Shear stress (psi)	Environment	Status
1	Standard Set	770	70°F, Dry	Stopped after 1000 hrs
2	Standard Set	770	70°F, Wet	Cohesive failure after 988 hours
3	Standard Set	770	110°F, Dry	Cohesive failure during loading
4	Standard Set	165	110°F, Dry	Stopped after 1000 hrs
5	Fast Set	770	70°F, Dry	Cohesive failure during loading
6	Fast Set	770	70°F, Wet	Bonding Failure during loading
7	Fast Set	770	110°F, Dry	Cohesive failure during loading
8	Fast Set	165	110°F, Dry	Cohesive failure during loading
9	Fast Set	385	70°F, Dry	Stopped after 1000 hrs
10	Fast Set	385	70°F, Wet	Cohesive failure after 4 hours
11	Fast Set	385	70°F, Dry	Cohesive failure during loading
12	Fast Set	192	70°F, Dry	Cohesive failure after 684 hours

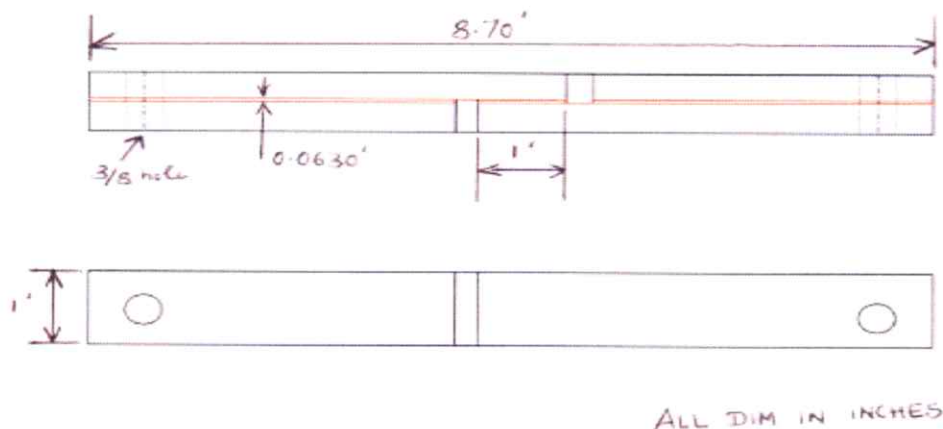


Figure 1 Test specimen geometry

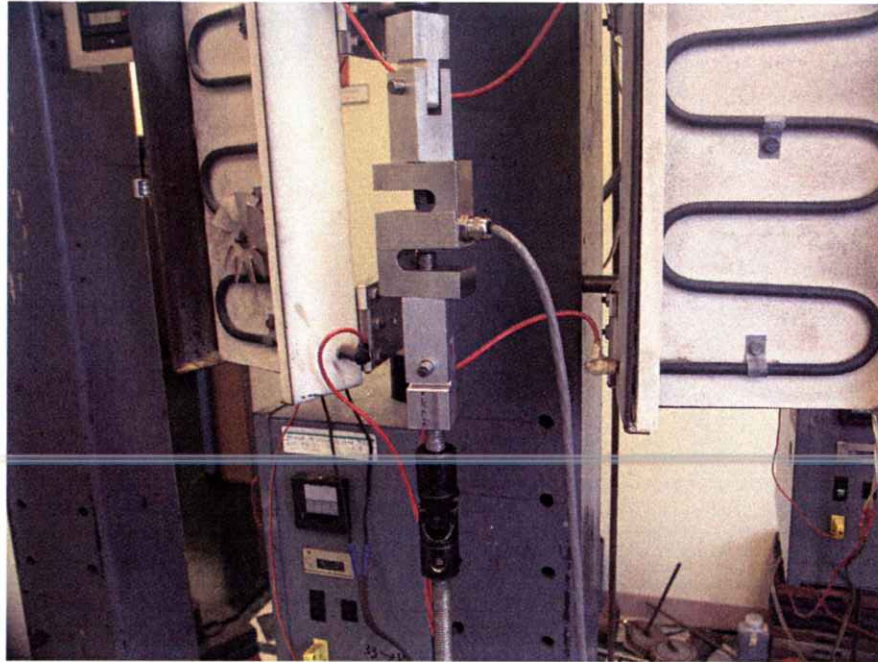


Figure 2 Load cell attached to the creep stand

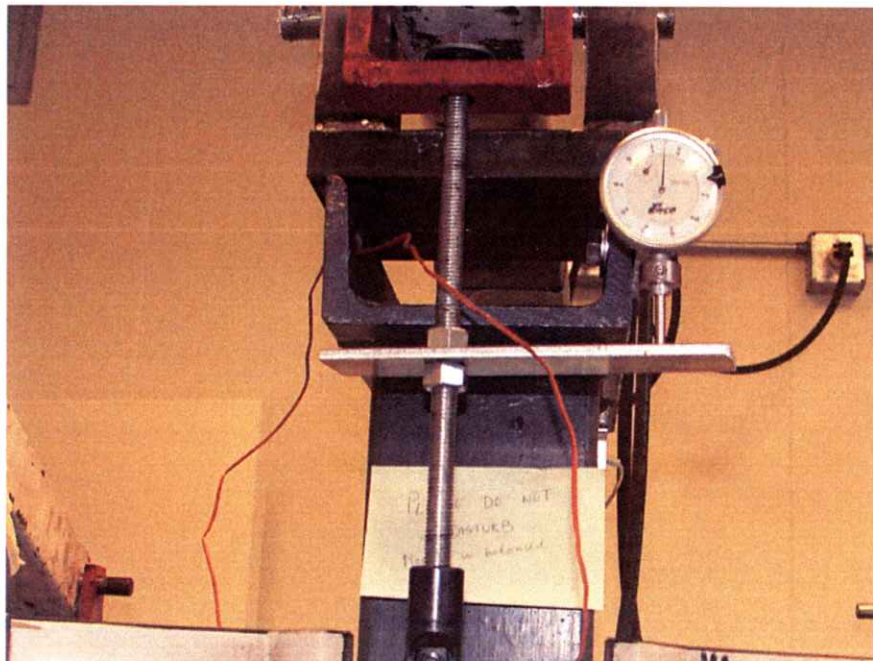


Figure 3 Dial gage attached to the creep stand

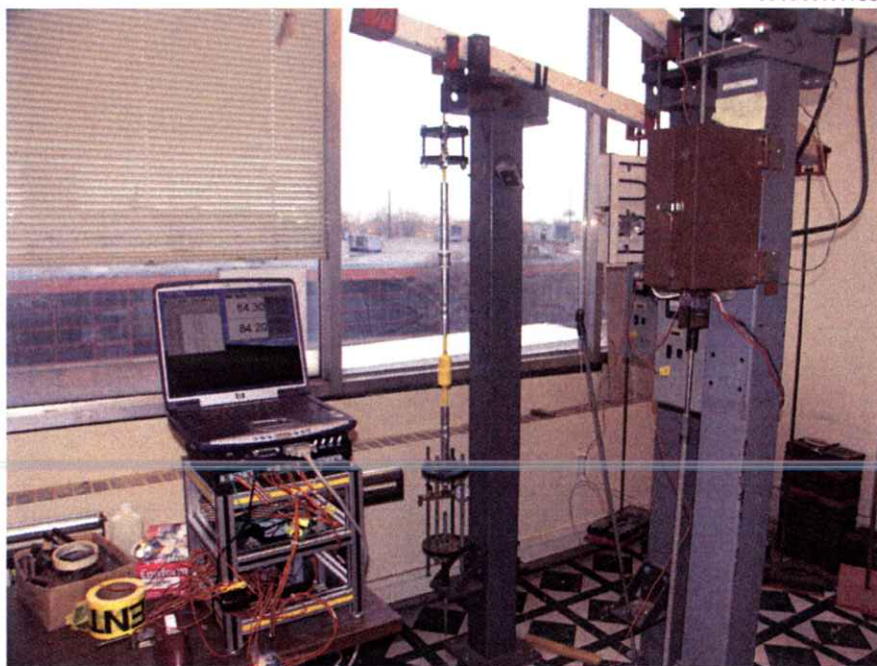


Figure 4 Heaters fitted in the creep stands and temperature data acquisition system

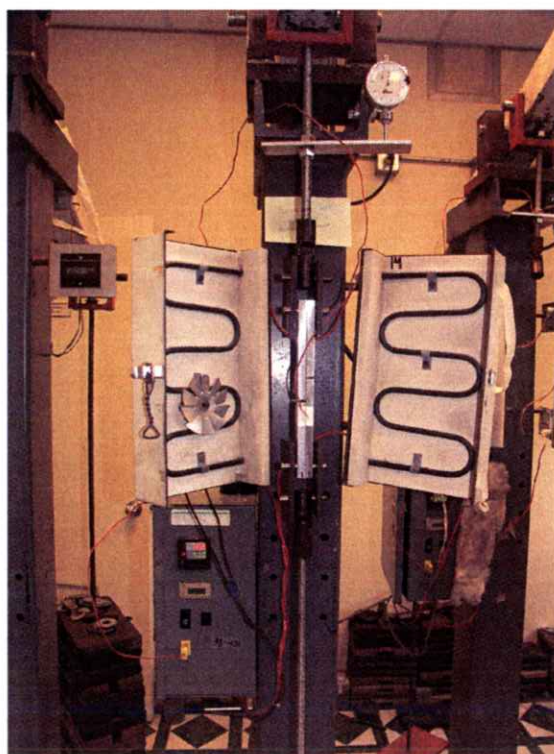


Figure 5 Lap shear creep test

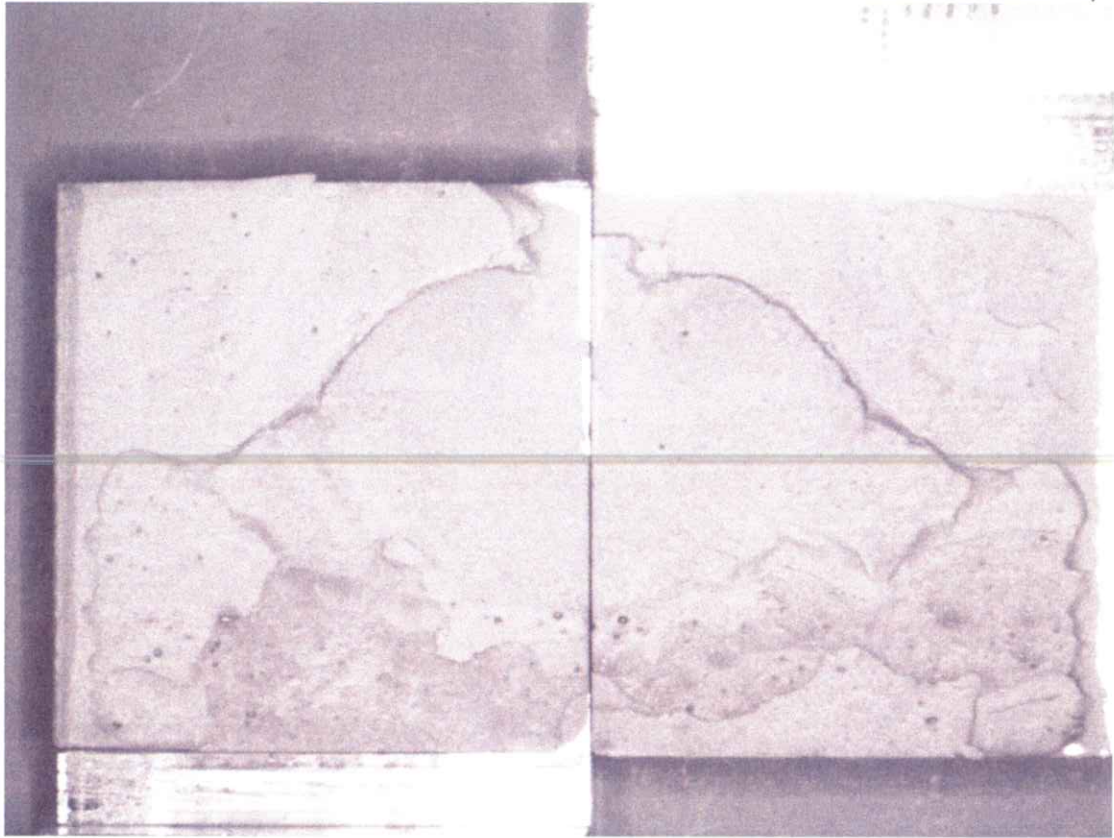


Figure 6 Failure surface of Standard set epoxy loaded at 770 Psi, 110°F, Dry condition (Sample 3)

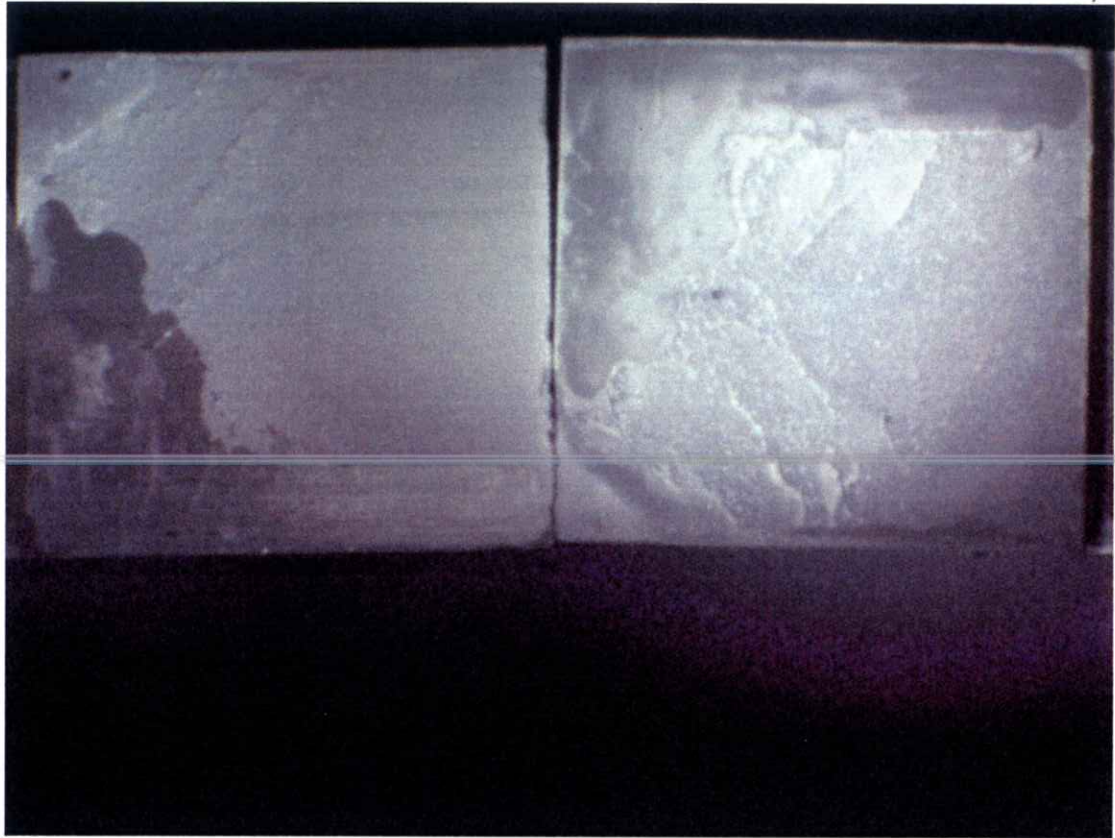
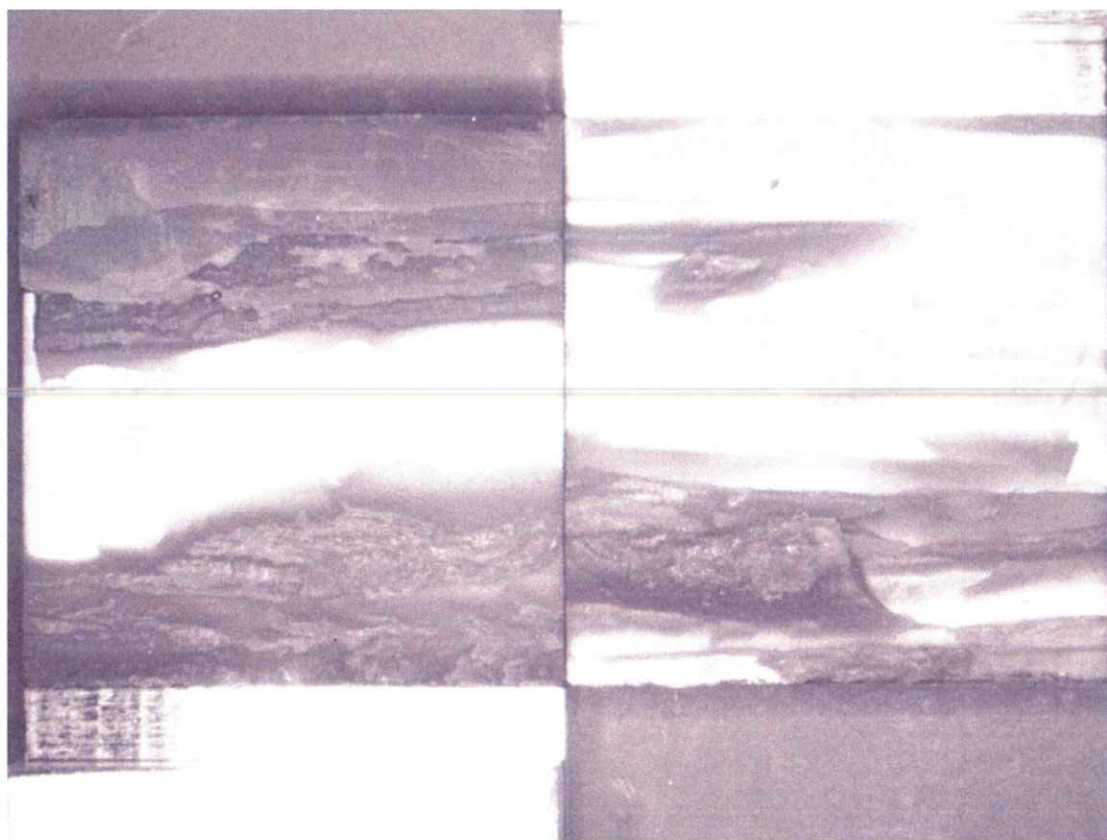
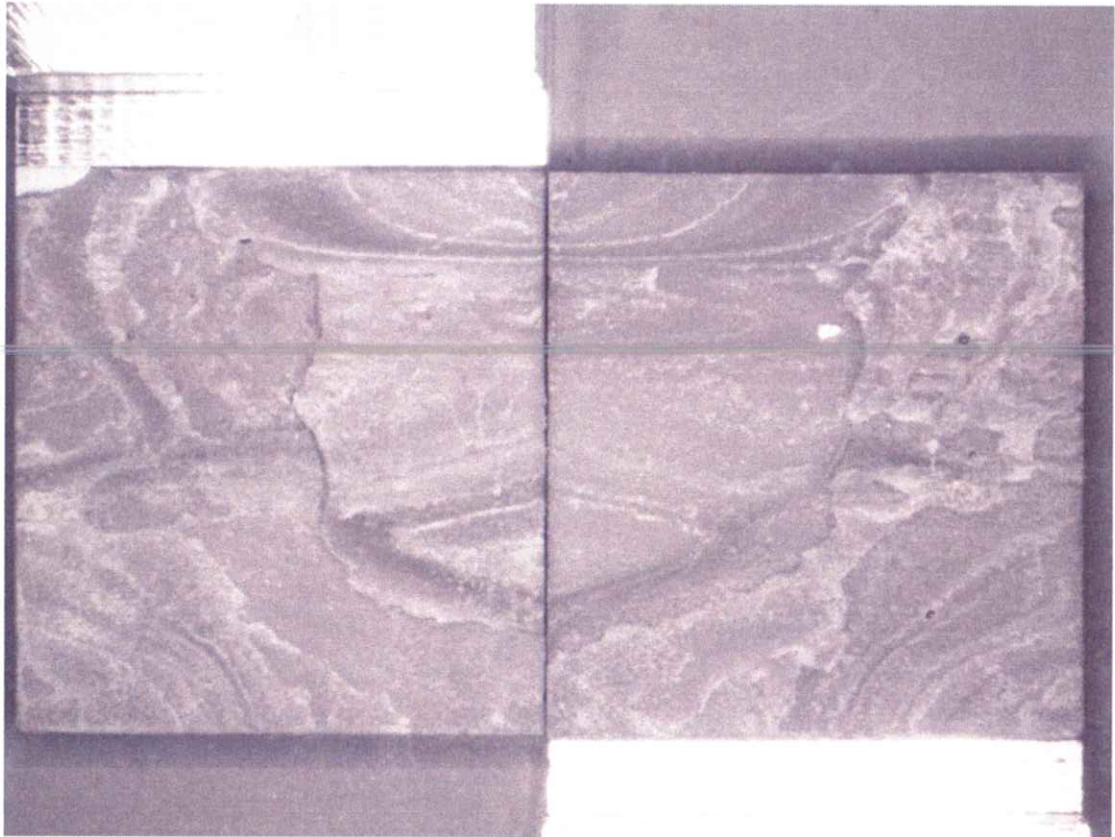


Figure 7 Failure surface of Standard set epoxy loaded at 770Psi, 70°F, Wet condition (Sample 2)



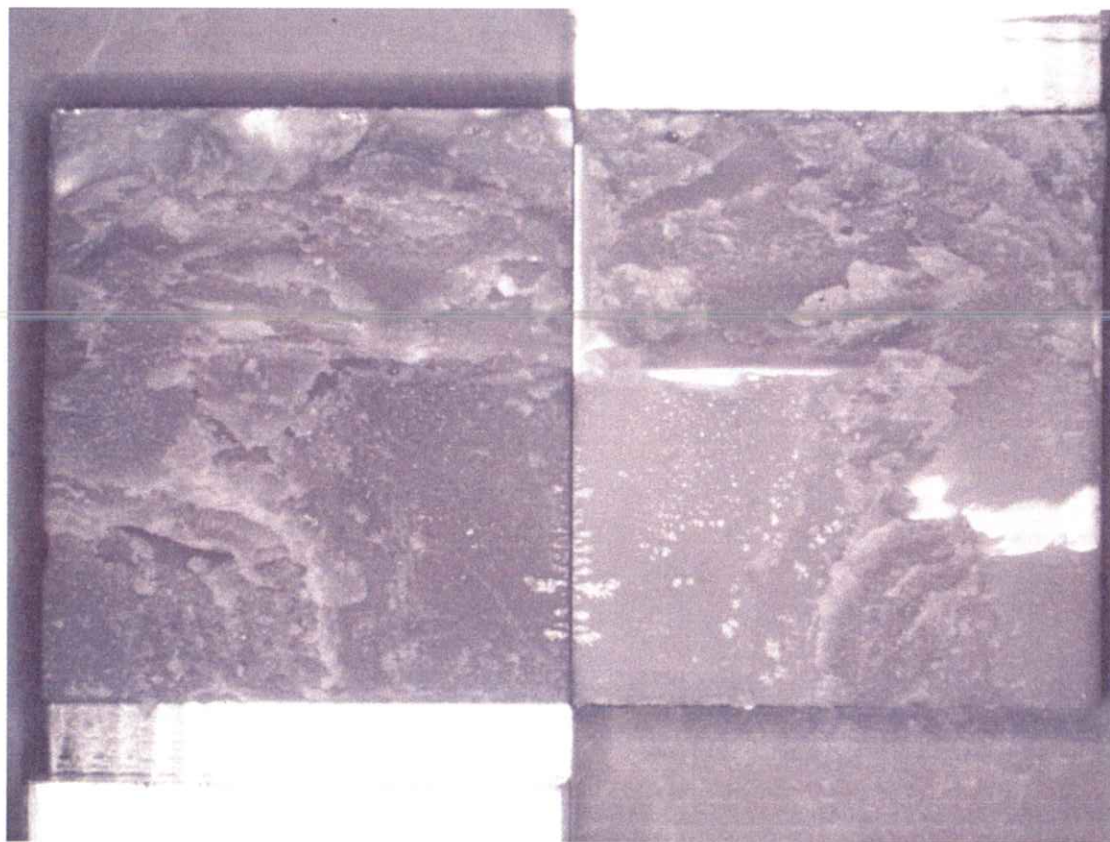
*Figure 8 Failure surface of Fast set epoxy loaded at 770 Psi, 70°F, Dry condition
(Sample 5)*



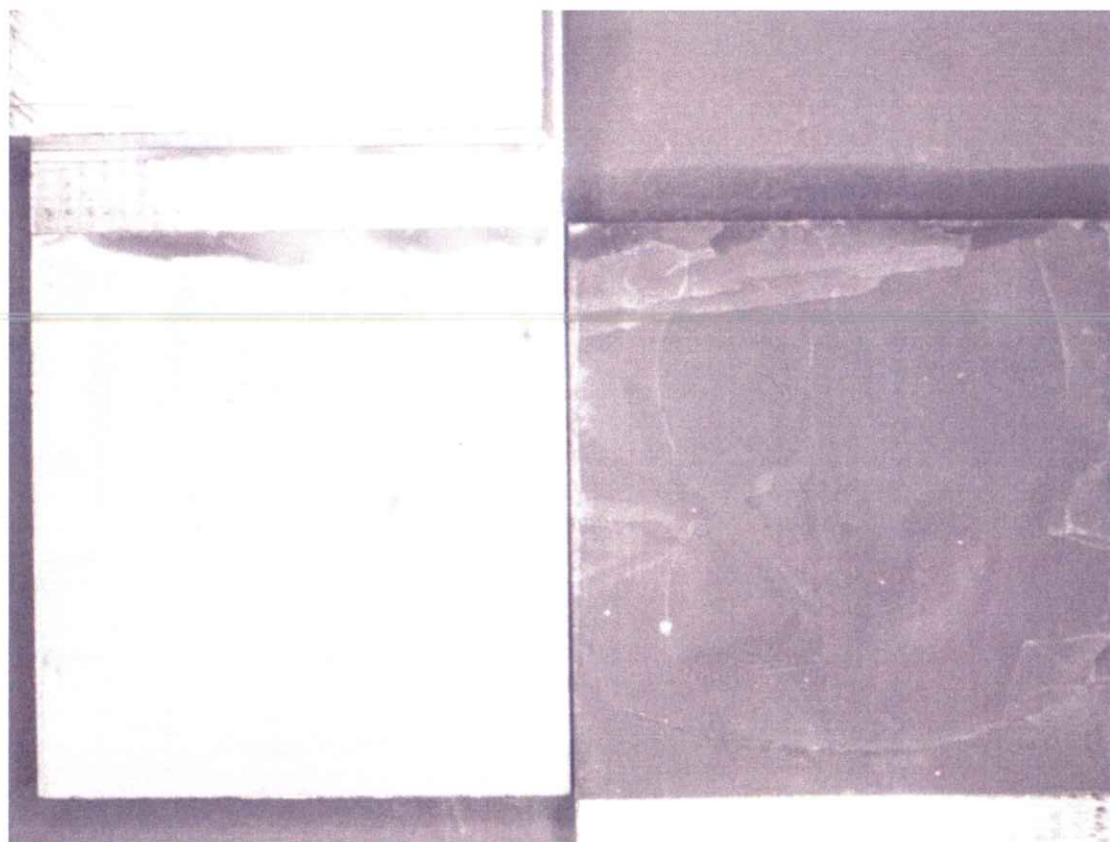
*Figure 9 Failure surface of Fast set epoxy loaded at 165 Psi, 110°F, Dry condition
(Sample 8)*



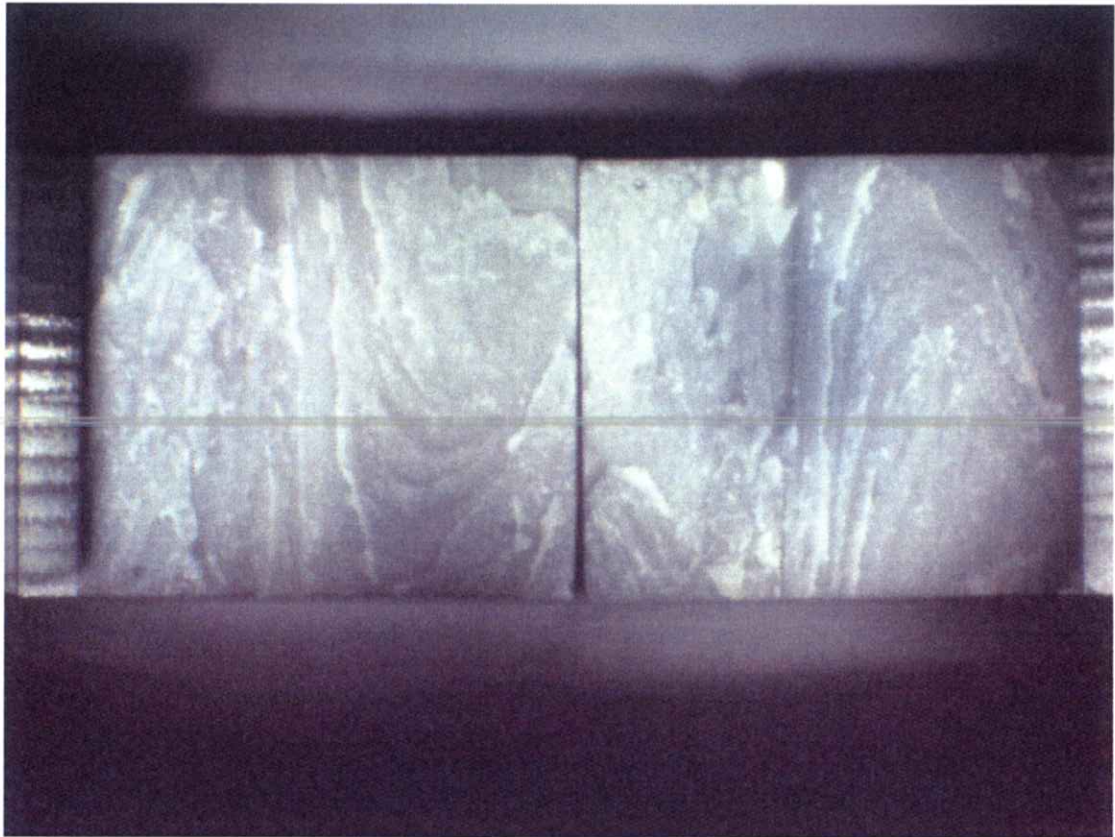
*Figure 10 Failure surface of Fast set epoxy loaded at 770 Psi, 110°F, Dry condition
(Sample 7)*



*Figure 11 Failure surface of Fast set epoxy loaded at 385Psi, 70°F, Dry condition
(Sample 11)*



*Figure 12 Failure surface of Fast set epoxy loaded at 385Psi, 70°F, Wet condition
(Sample 10)*



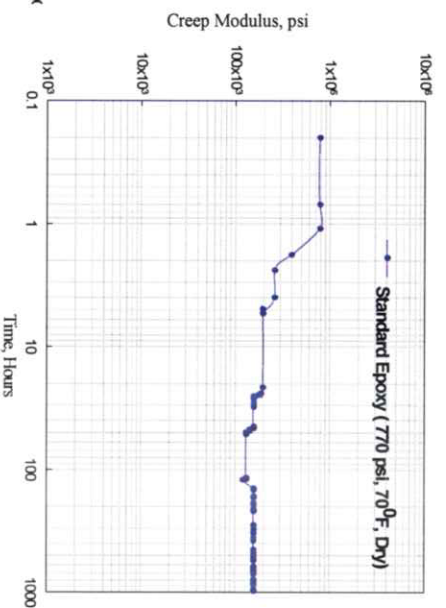
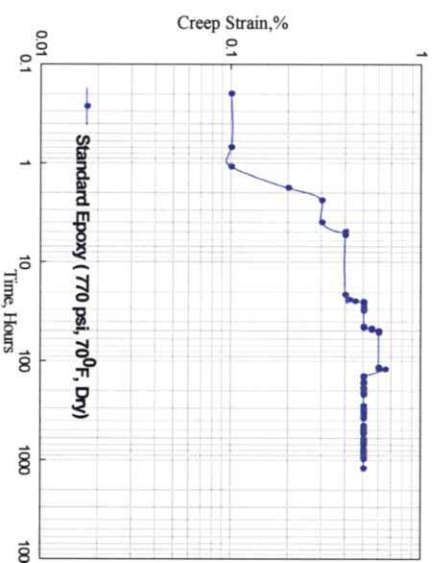
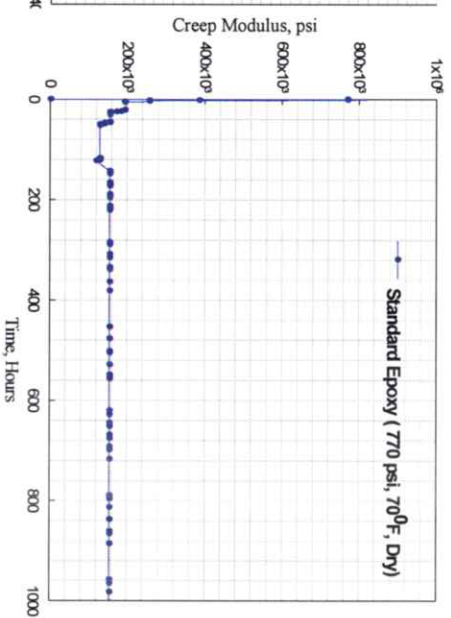
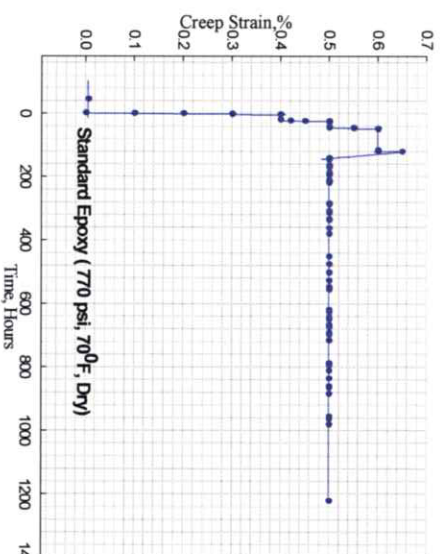
*Figure 13 Failure surface of Fast set epoxy loaded at 192Psi, 70°F, Dry condition-
(Sample 12)*

APPENDIX



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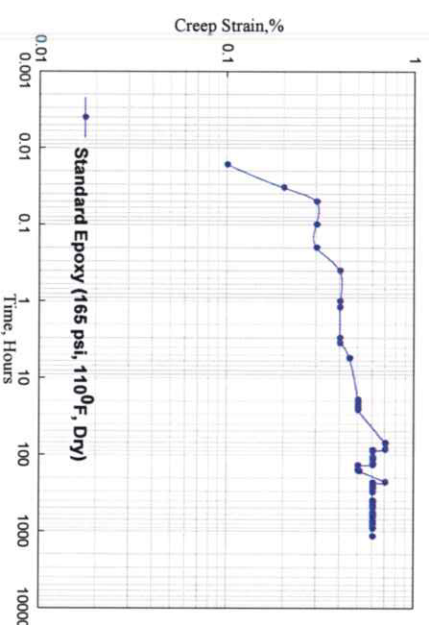
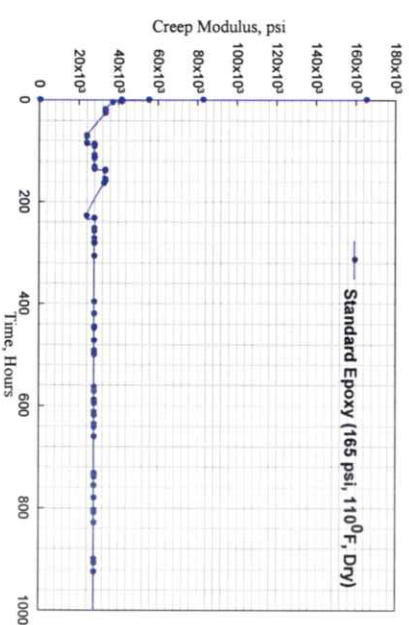
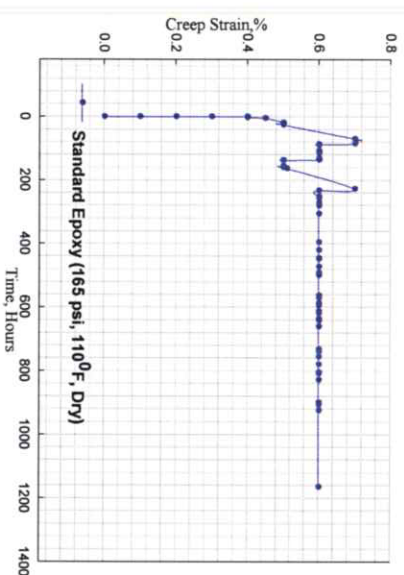
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Stress Level: 770 psi
Environment: 70°F, Dry
Tested by: Dr. Ahamed Shabeer
Signature: _____
Reviewed by: _____





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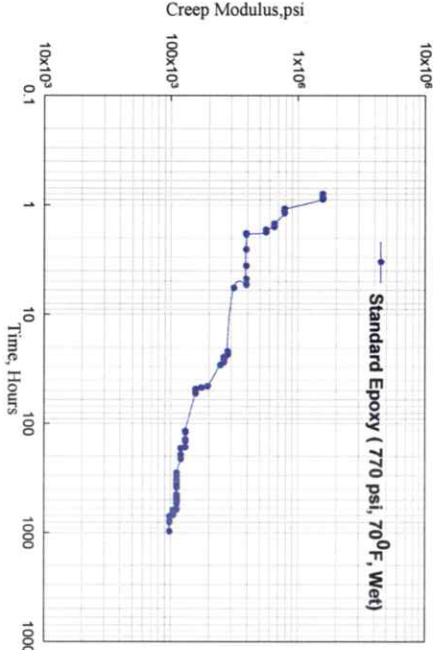
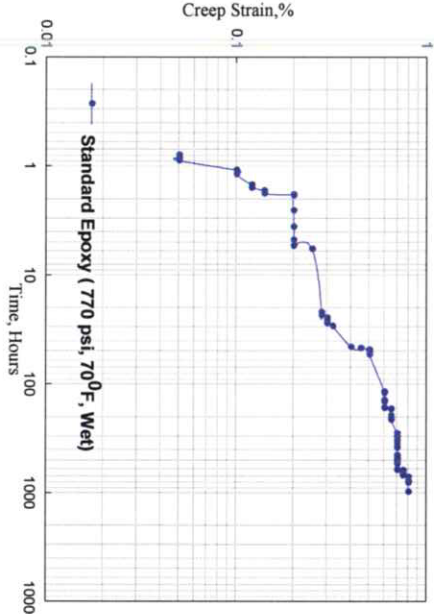
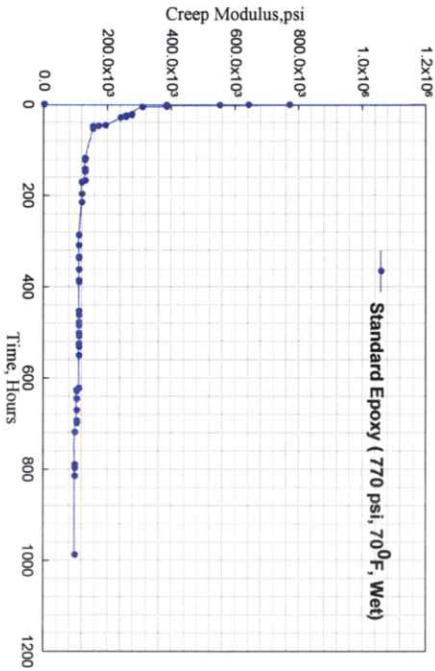
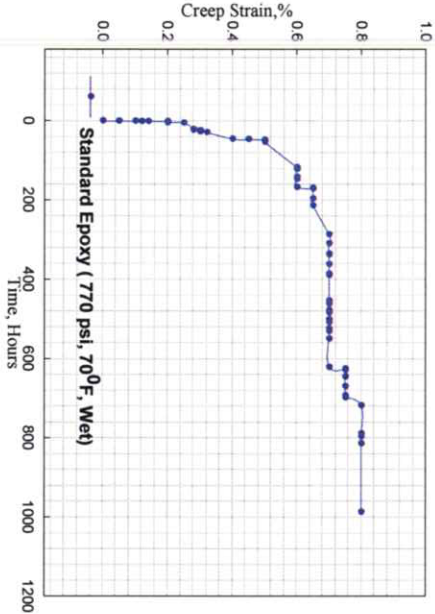
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Stress Level: 165 psi
Environment: 110°F, Dry
Tested by: Dr. Ahamed Shabeer
Signature: _____
Reviewed by: _____





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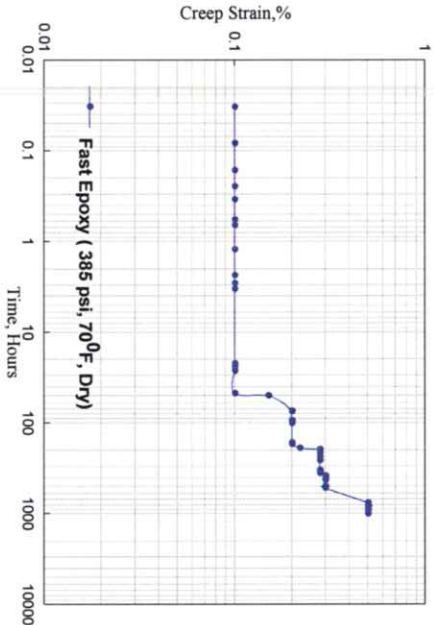
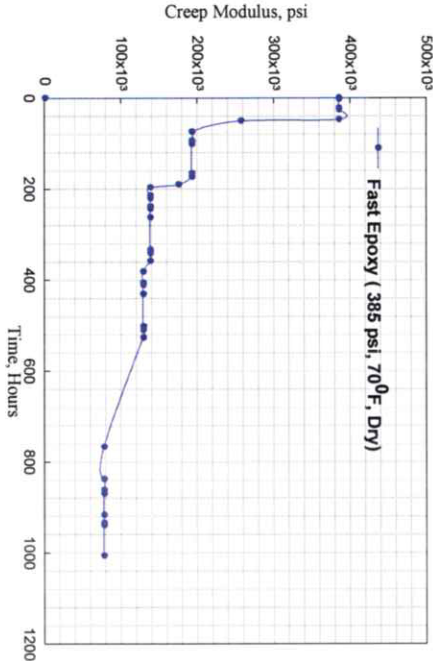
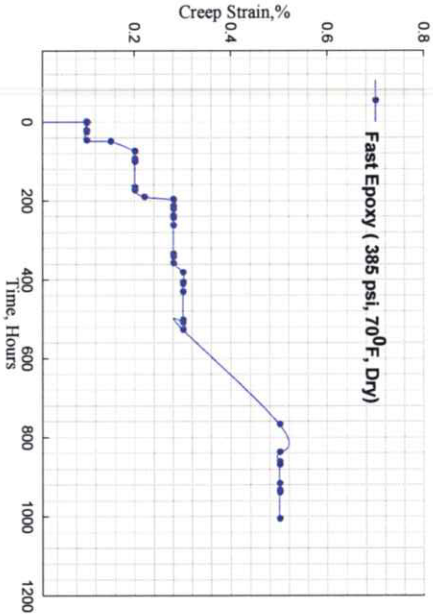
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Stress Level: 770 psi
Environment: 70°F, Wet
Tested by: Dr. Ahamed Shabeer
Signature: _____
Reviewed by: _____





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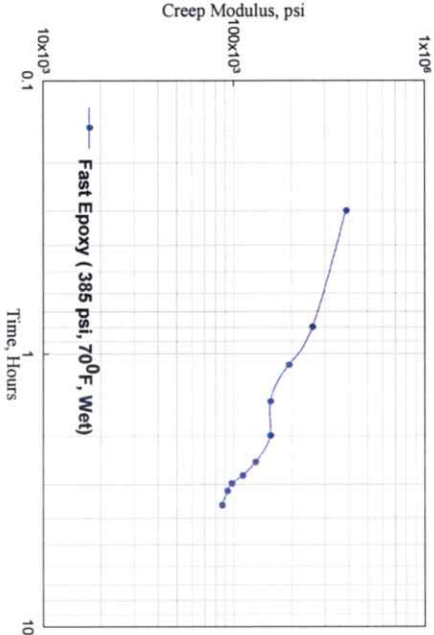
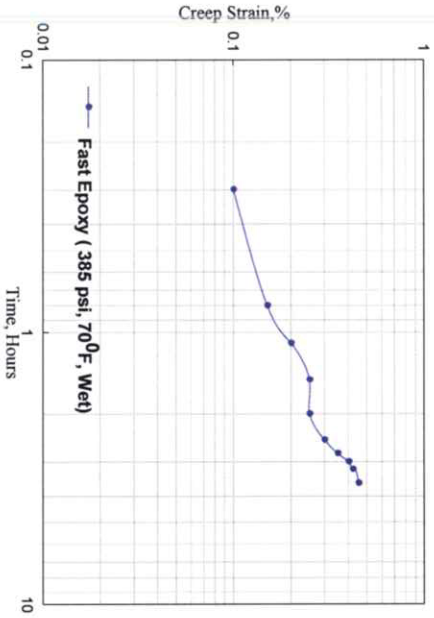
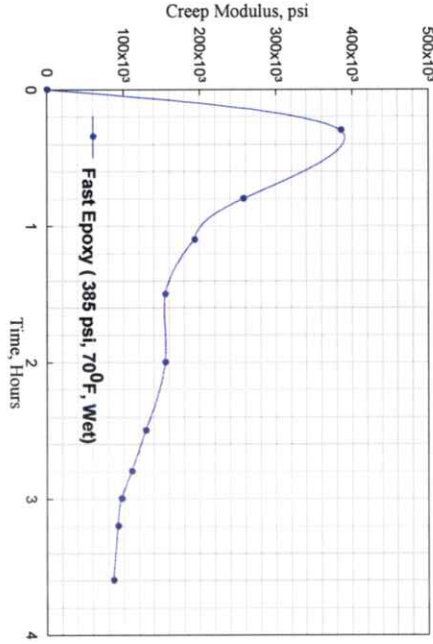
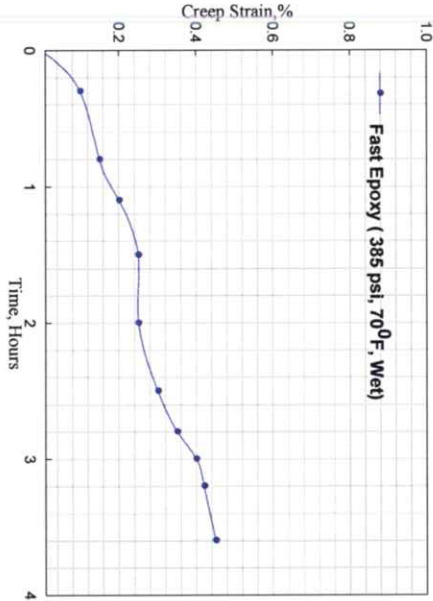
Material: Fast Set Epoxy Adhesive
Stress Level: 385 psi
Environment: 70°F, Dry
Tested by: Dr. Ahmed Shabeer
Signature: _____
Reviewed by: _____





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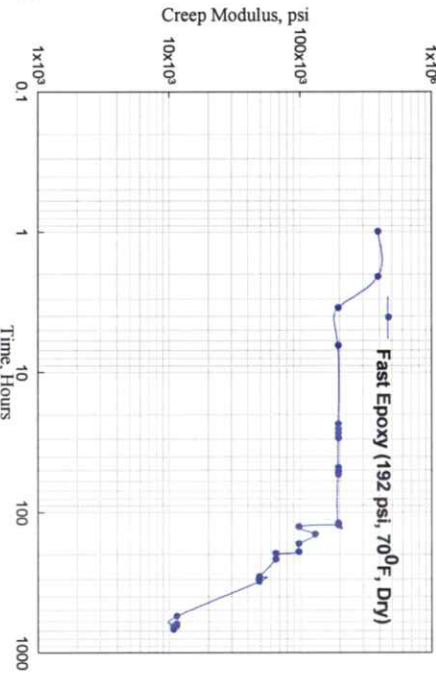
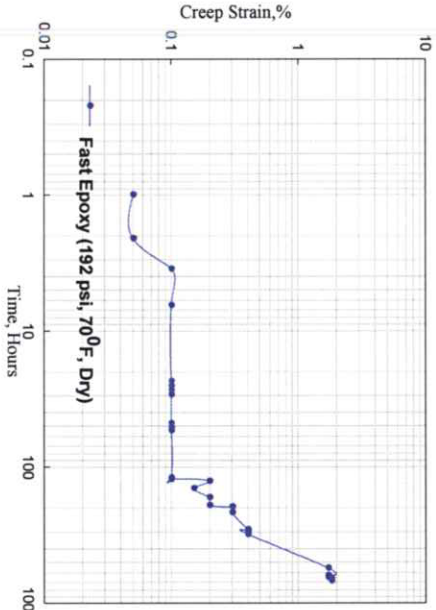
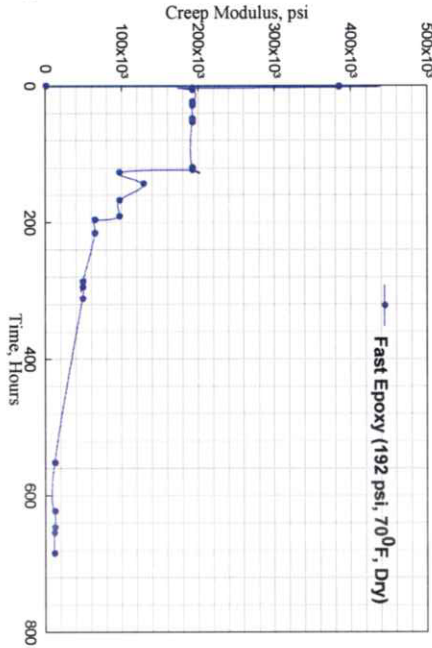
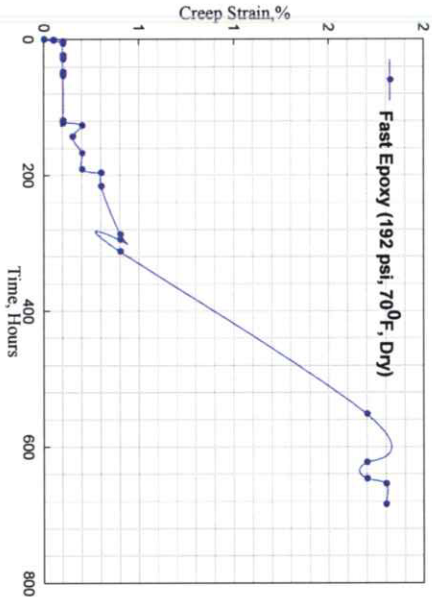
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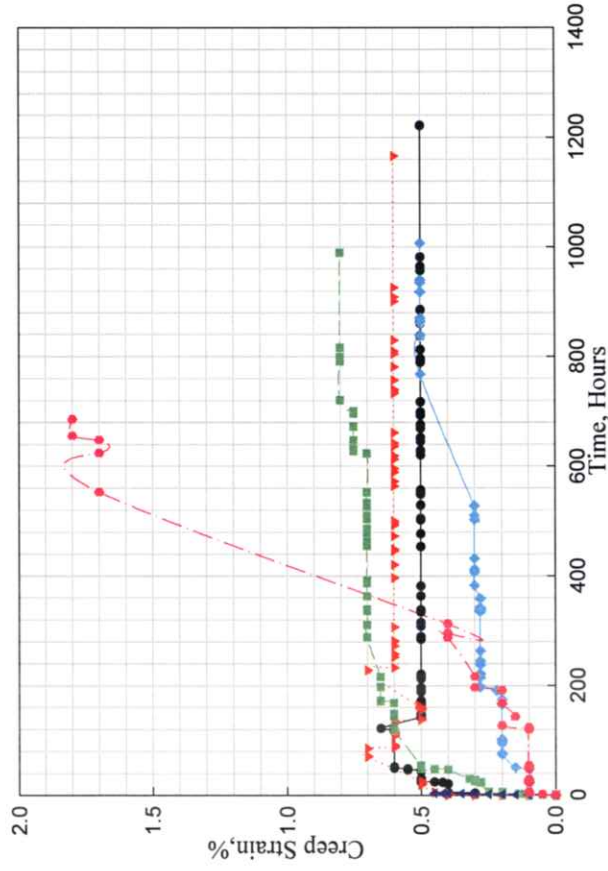




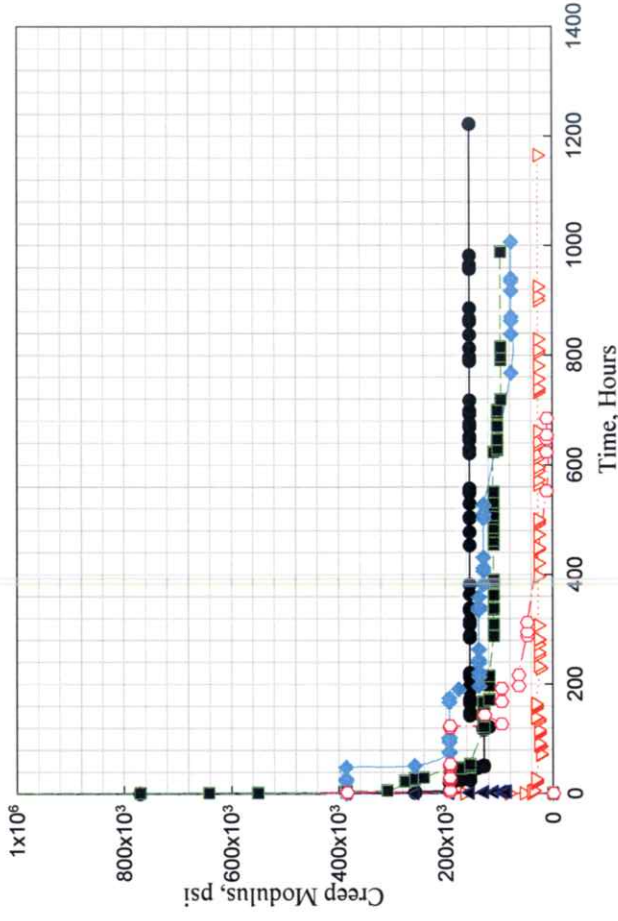
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www.mtusa.bodycote.com

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Stress Level: 192 psi
Environment: 70°F, Dry
Tested by: Dr. Ahmed Shabeer
Signature: 
Reviewed by: 





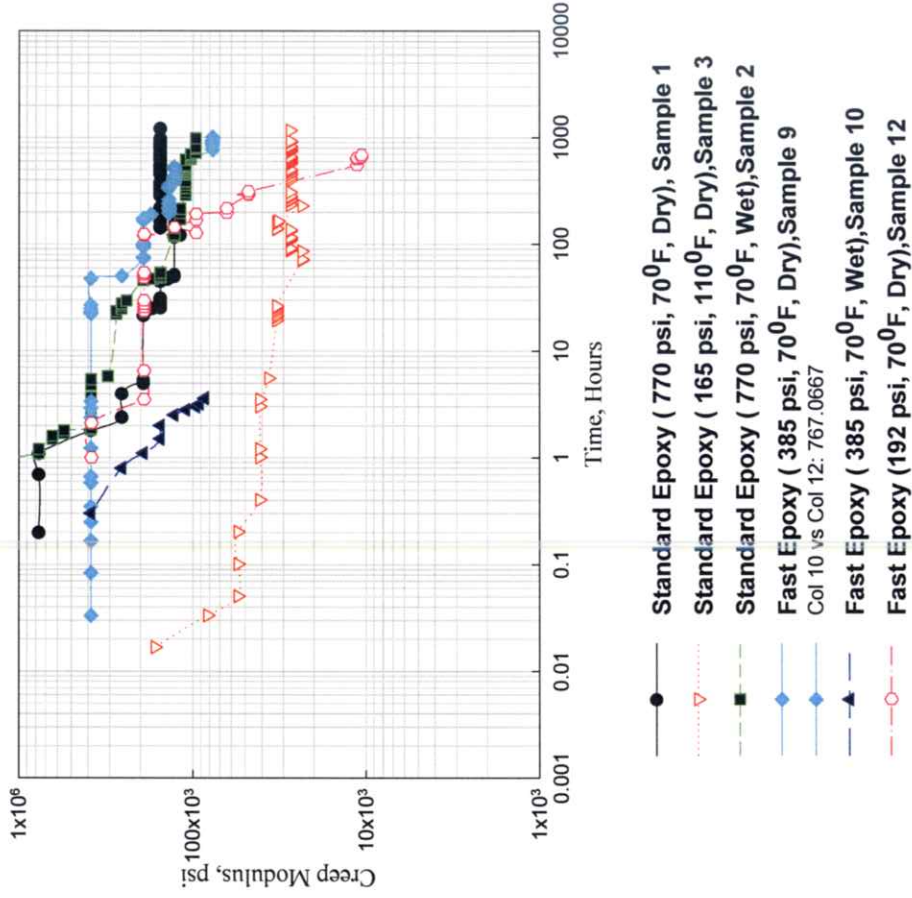
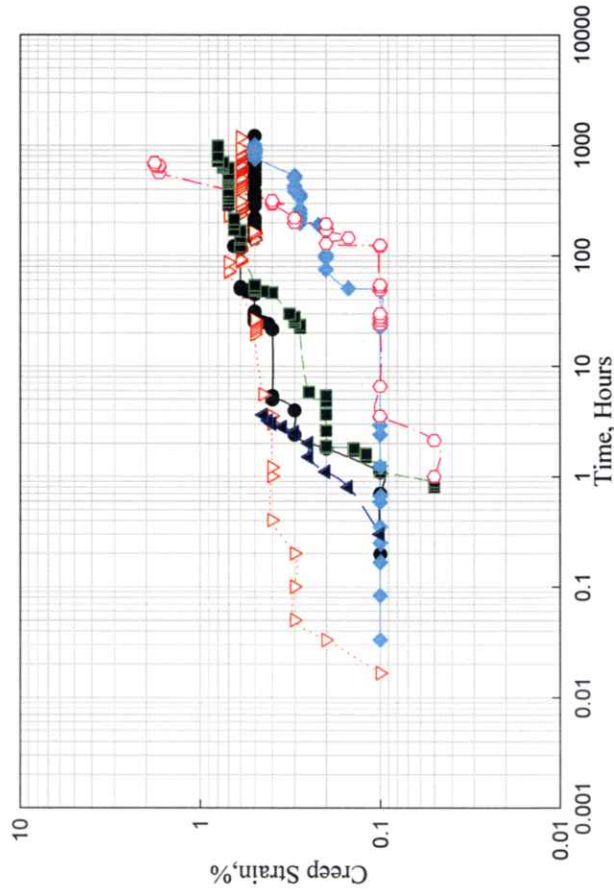
- Standard Epoxy (770 psi, 70⁰F, Dry), Sample 1
- △— Standard Epoxy (165 psi, 110⁰F, Dry), Sample 3
- Standard Epoxy (770 psi, 70⁰F, Wet), Sample 2
- ◆— Fast Epoxy (385 psi, 70⁰F, Dry), Sample 9
- ▲— Fast Epoxy (385 psi, 70⁰F, Wet), Sample 10
- Fast Epoxy (192 psi, 70⁰F, Dry), Sample 12



- Standard Epoxy (770 psi, 70⁰F, Dry), Sample 1
- △— Standard Epoxy (165 psi, 110⁰F, Dry), Sample 3
- Standard Epoxy (770 psi, 70⁰F, Wet), Sample 2
- ◆— Fast Epoxy (385 psi, 70⁰F, Dry), Sample 9
- ▲— Fast Epoxy (385 psi, 70⁰F, Wet), Sample 10
- Fast Epoxy (192 psi, 70⁰F, Dry), Sample 12



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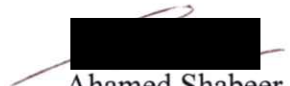
30003712

NTSB

Lap Shear Strength Testing of Epoxy Adhesives

Date: 28th March 2007

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INDEX

REPORT

APPENDIX

Introduction

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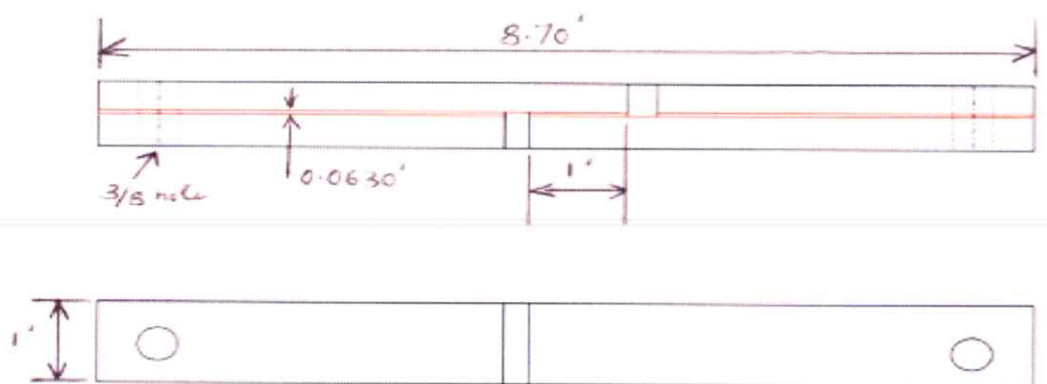
Sample Preparation

Test specimens of the required dimensions were prepared. Rectangular sections of aluminum were milled to the required length. The aluminum blanks (6061) and the shims were then cleaned using trichloroethane to remove any grease. Then the specimens were washed with deionized water. After it is dried, BR 127 primer was applied to all sides of the specimen. Care was taken for those specimens in which the creep tests were conducted in wet conditions. Then the primer was cured at 250°F for 30 minutes. The aluminum sections were bonded using the epoxy adhesives and allowed to cure as per the data sheet supplied by the manufacturer. After the specimens have fully cured, they are machined to get the required dimensions. The test specimen geometry is shown in Figure 1.

Testing

Lap shear Strength per ASTM D5656

The testing was conducted according to ASTM D5656 “Standard Test Method for Thick-Adherend Metal Lap-Shear Joints for Determination of the Stress-Strain Behavior of Adhesives in Shear by Tension Loading “at a crosshead displacement speed of 0.015 inches/minute using an Instron 4505 universal testing machine. Three specimens were tested at room temperature. The test specimens were conditioned at the test temperature for a minimum of 40 hours prior to testing. The strength was measured and the average and standard deviation values are reported in Appendix. The stress versus extension curves and the specimen results are reported in the Appendix.



ALL DIM IN INCHES

Figure 1 Test specimen geometry

APPENDIX



Test Certificate

3/27/07

Sample ID: 30003712- NTSB
Method: ASTM D5656 Shear by Tension Loading

Test Date: 3/27/07
Operator: AHAMED SHABEER

Sample Information:

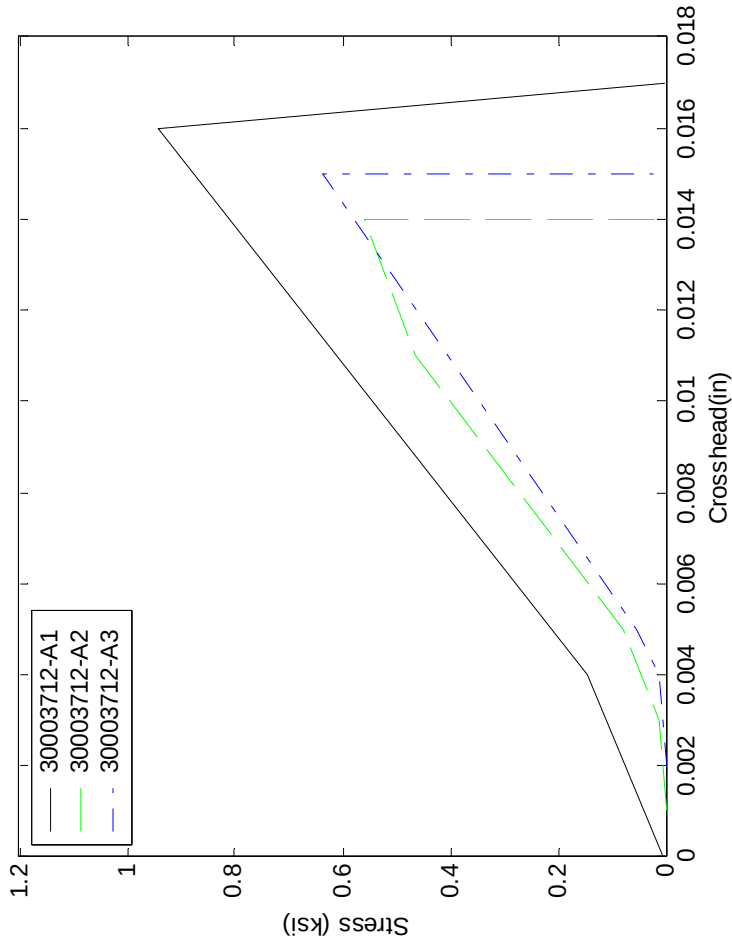
Name	Value
01) Client:	NTSB
02) BBI #:	30003712
03) Adhesive:	Fast set Epoxy
04) Adhenend Material:	Aluminum
05) Conditioning:	40hr@ 72°F & 50% humidity
06) Test Conditions:	72°F, 51% RH
07) Test Method:	ASTM D5656-01

Specimen Results:

Specimen I.D.#	Type	Width (in)	Length (in)	Peak Load (lbf)	Peak Stress (ksi)
30003712-A1	Fast set	0.9680	1.0050	917.7	0.9
30003712-A2	Fast set	0.9520	1.0200	542.9	0.56
30003712-A3	Fast set	0.9560	1.0120	615.7	0.6
Mean		0.958667	1.012333	692.1	0.686667
Std. Dev.		0.008327	0.007506	198.7372	0.185831

Test Inputs:

Name	Value	Units
Break Sensitivity	90	%
Break Threshold	0.500	lbf
Data Acq. Rate	10.0	Hz
Test Speed	0.015	in/min



Sample ID: 30003712- NTSB

Method: ASTM D5656 Shear by Tension Loading.

Test performed by: Dr. Ahamed Shabeer



Test Certificate

3/27/07

Sample ID: 30003712- NTSB
Method: ASTM D5656 Shear by Tension Loading

Test Date: 3/27/07
Operator: AHAMED SHABEER

Sample Information:

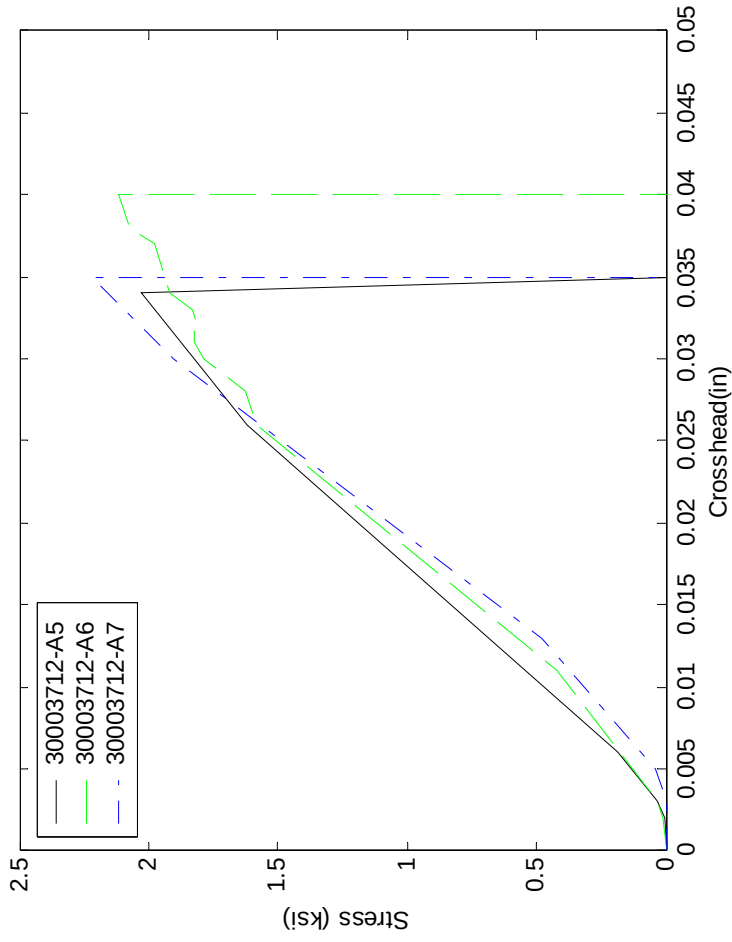
Name	Value
01) Client:	NTSB
02) BBI #:	30003712
03) Adhesive:	Standard set Epoxy
04) Adhenead Material:	Aluminum
05) Conditioning:	40hr@ 72°F & 50% humidity
06) Test Conditions:	72°F, 51% RH
07) Test Method:	ASTM D5656-01

Specimen Results:

Specimen I.D.#	Type	Width (in)	Length (in)	Peak Load (lbf)	Peak Stress (ksi)
30003712-A5	Standard set	0.9290	1.0710	2015.8	2.0
30003712-A6	Standard set	0.9480	1.1012	2207.9	2.1
30003712-A7	Standard set	0.9540	1.0050	2123.3	2.2
Mean		0.943667	1.059067	2115.667	2.1
Std. Dev.		0.013051	0.049198	96.27722	0.1

Test Inputs:

Name	Value	Units
Break Sensitivity	90	%
Break Threshold	0.500	lbf
Data Acq. Rate	10.0	Hz
Test Speed	0.015	in/min



Sample ID: 30003712- NTSB

Method: ASTM D5656 Shear by Tension Loading.

Test performed by: Dr. Ahamed Shabeer